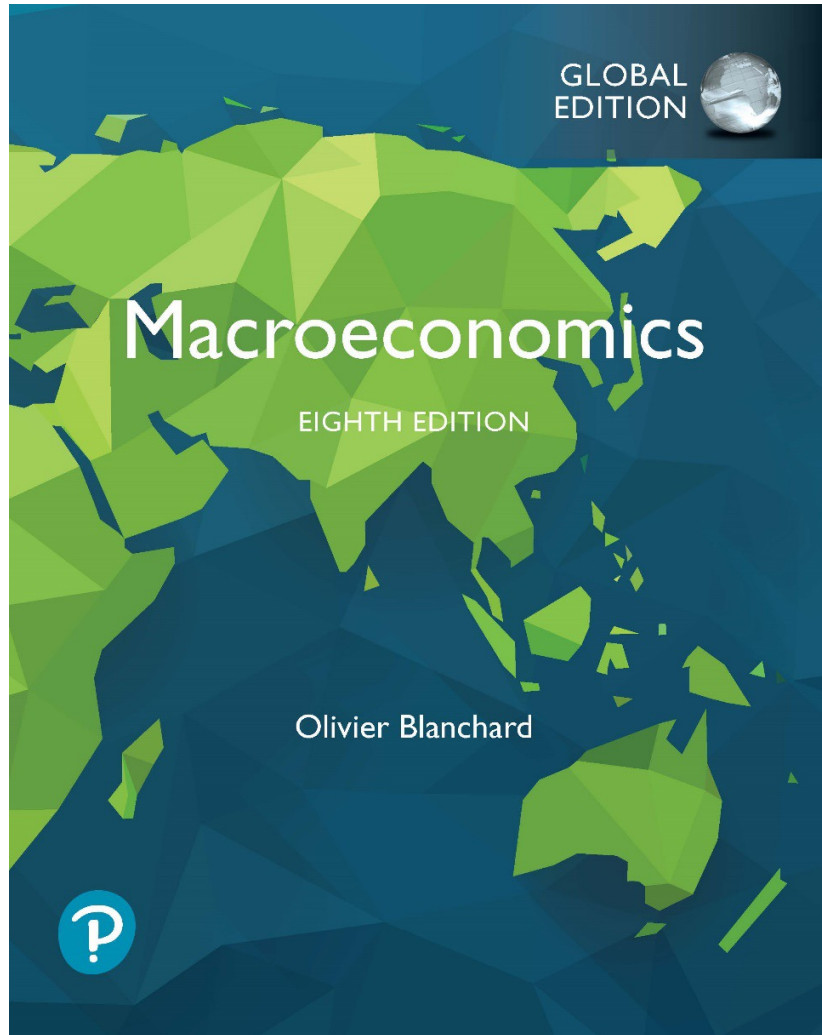


Macroeconomics

Eighth Edition, Global Edition



Chapter 4

Financial Markets I

Chapter 4 Outline

Financial Markets I

4.1 The Demand for Money

4.2 Determining the Interest Rate: I

4.3 Determining the Interest Rate: II

4.4 The Liquidity Trap

Financial Markets

- Financial markets are intimidating, but they play an essential role in the economy.
- In this chapter, we focus on the role of the central bank in affecting these interest rates.
- We learn how the interest rate on bonds is determined, and the role of the central bank (**Federal Reserve Bank**, or the **Fed**, in the United States) in this determination.

$$\begin{array}{l} W = W^H + W^F \\ \text{(wealth)} = M + B \end{array}$$

↑ human
↓ financial

財富為一存量 (stock)
收入... 流量 (flow)

$$\begin{aligned} M^d + B^d &= M^s + B^s \\ M^d - M^s &= B^s - B^d \\ \underline{M^d = M^s} &\Leftrightarrow \underline{B^s = B^d} \end{aligned}$$

eqm. eqm.

4.1 The Demand for Money (1 of 5)

- Suppose you only have a choice between two assets: money and bonds.
- *Money* are used for transactions, but it pays no interest.
- Two types of money: **currency** and **checkable deposits**.
- **Bonds** pay a positive interest rate, i (*the* rate of interest), but cannot be used for transactions.

4.1 The Demand for Money (2 of 5)

- The holding of money and bonds depends on:
 - Your level of transactions
 - The interest rate on bonds
- You can hold bonds indirectly through **money market funds**, or *money market mutual funds*.
- In the early 1980s, the interest rate on money market funds reached 14% per year, so people earned more interest by moving their wealth from checking accounts to these funds.

FOCUS: Semantic Traps: Money, Income, and Wealth

- **Money** is what can be used to pay for transactions.
- **Income** is what you earn, and it is a flow.
- **Saving** is the part of after-tax income that you do not spend, and it is also a flow.
- **Savings** is the value of what you have accumulated over time.
- **Financial wealth**, or wealth, is the value of all your financial assets minus all your financial liabilities, and it is a stock variable.
- **Investment** is what economists refer to as the purchase of new capital goods.
- **Financial investment** is the purchase of shares or other financial assets.

4.1 The Demand for Money (3 of 5)

- Demand for money (M^d) is equal to nominal income $\$Y$ (a measure of level of transactions in the economy) times a decreasing function of the interest rate i :

$$M^d = \$Y L(i) \quad (4.1)$$

- An increase in the interest rate decreases the demand for money, as people put more of their wealth into bonds.

4.1 The Demand for Money (4 of 5)

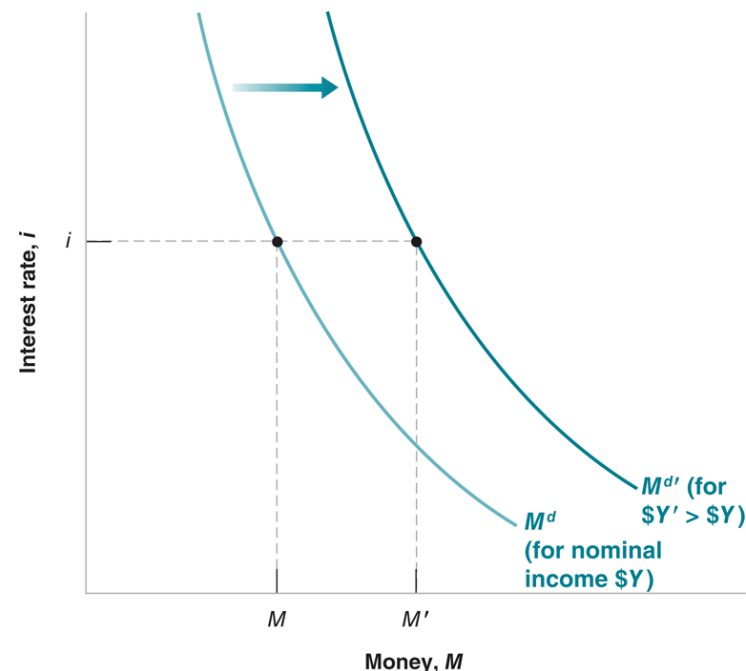
- Equation (4.1) means that the demand for money:
 - increases in proportion to nominal income, and
 - depends negatively on the interest rate.
- The relation between the demand for money and interest rate for a given level of income Y is represented by the M^d curve.

4.1 The Demand for Money (5 of 5)

Figure 4.1 The Demand for Money

For a given level of nominal income, a lower interest rate increases the demand for money.

At a given interest rate, an increase in nominal income shifts the demand for money to the right.



FOCUS: Who Holds U.S. Currency

- The amount of currency in circulation in 2006 was \$750 billion.
- U.S. households together held \$170 billion in currency.
- U.S. firms held another \$80 billion.
- Foreigners abroad held \$500 billion, or 66% of the total, for transactions, especially in countries suffering from high inflation in the past.

4.2 Determining the Interest Rate: I (1 of 9)

- Suppose the central bank decides to supply an amount of money equal to M :

$$M^s = M$$

- Equilibrium in financial markets requires that $M^s = M^d = M$:

Bonds market also be eqm.

Money supply = Money demand

$$M = \cancel{\$Y} L(i)$$

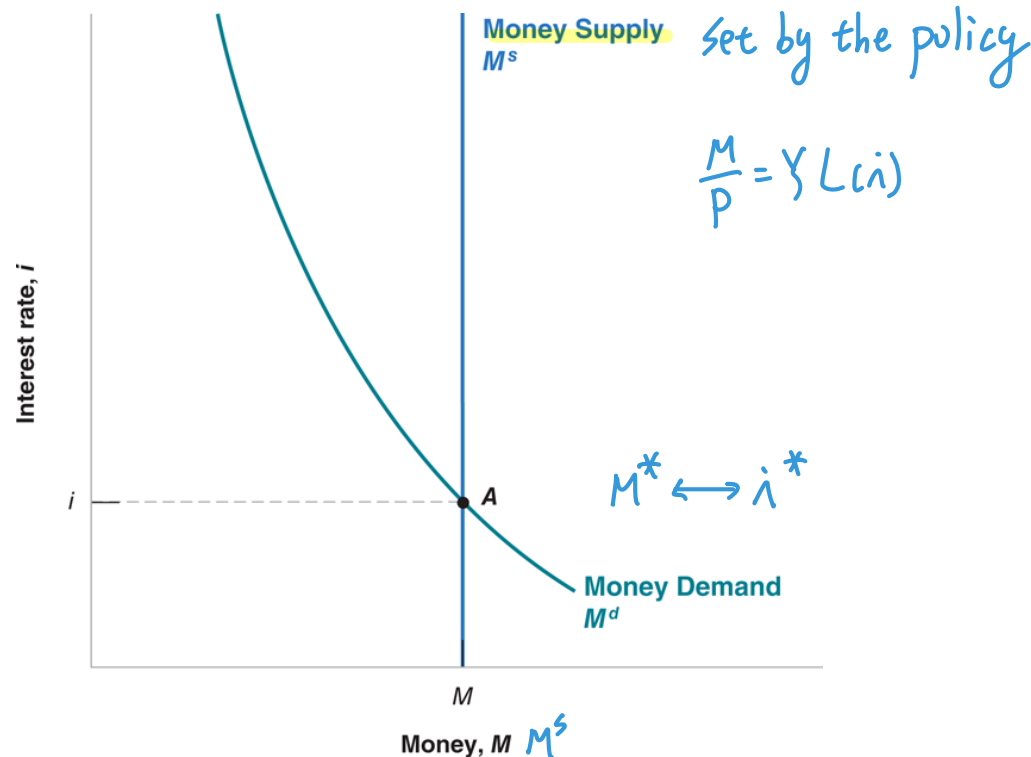
(4.2)

外生变量

4.2 Determining the Interest Rate: I (2 of 9)

Figure 4.2 The Determination of the Interest Rate

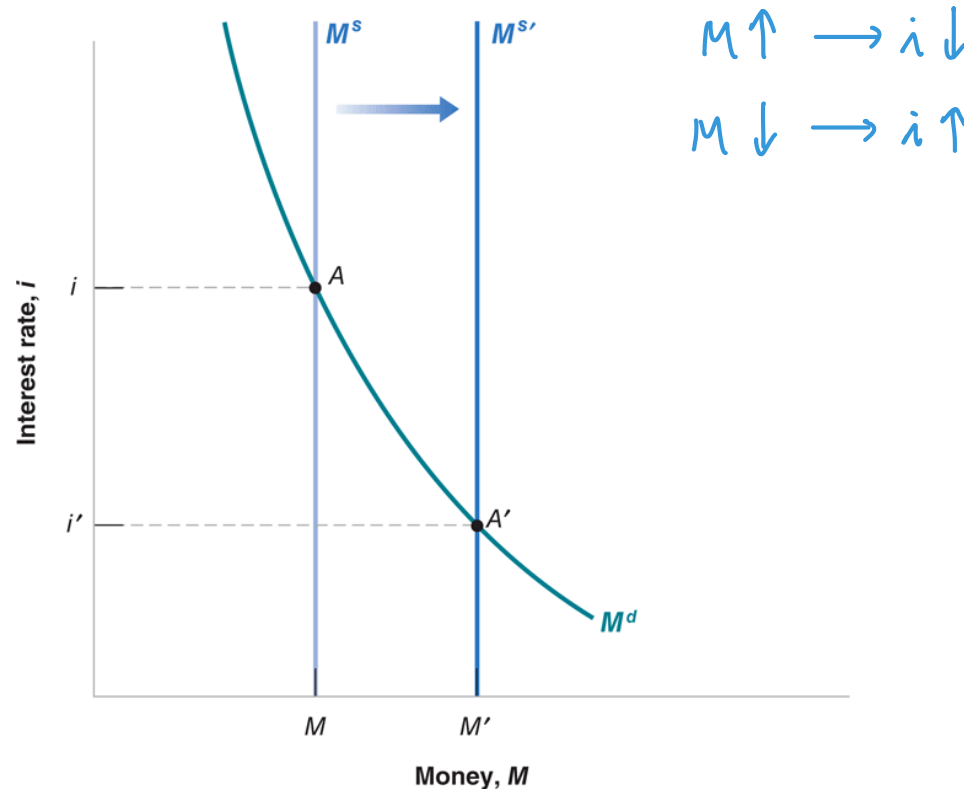
The interest rate must be such that the supply of money (which is independent of the interest rate) is equal to the demand for money (which does depend on the interest rate).



4.2 Determining the Interest Rate: I (3 of 9)

Figure 4.3 The Effects of an Increase in the Money Supply on the Interest Rate

An increase in the supply of money leads to a decrease in the interest rate.



4.2 Determining the Interest Rate: I (4 of 9)

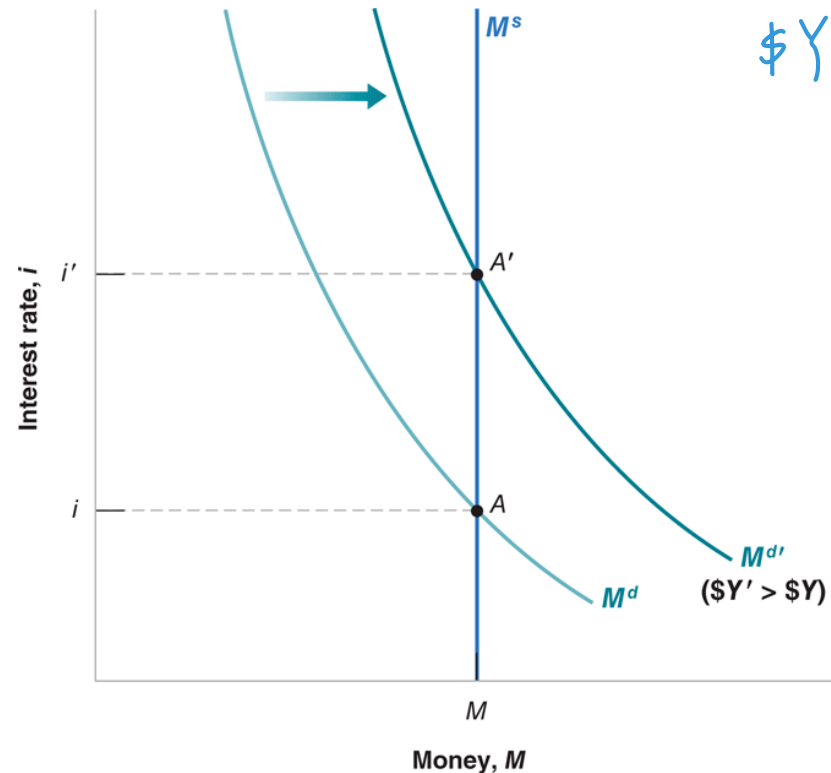
Figure 4.4 The Effects of an Increase in Nominal Income on the Interest Rate

Given the money supply, an increase in nominal income leads to an increase in the interest rate.

$$M = PYL(i)$$

$$PY \uparrow \rightarrow i \uparrow$$

↓ ↓



4.2 Determining the Interest Rate: I (5 of 9)

- For a given money supply, *an increase in nominal income leads to an increase in the interest rate.*
- *An increase in the supply of money by the central bank leads to a decrease in the interest rate.*

4.2 Determining the Interest Rate: I (6 of 9)

- Central banks typically change the supply of money by buying or selling bonds in the bond market—**open market operations**.
- **Expansionary open market operation:** the central bank *expands* the supply of money by buying bonds.
- **Contractionary open market operation:** the central bank *contracts* the supply of money by selling bonds.

增加錢的供給

$i \downarrow$

減少錢的供給

$i \uparrow$

4.2 Determining the Interest Rate: I (7 of 9)

Figure 4.5 The Balance Sheet of the Central Bank and the Effects of an Expansionary Open Market Operation

The assets of the central bank are the bonds it holds.

The liabilities are the stock of money in the economy.

An open market operation in which the central bank buys bonds and issues money increases both assets and liabilities by the same amount.

Balance Sheet	
Assets	Liabilities
Bonds	Money (currency)

The Effects of an Expansionary Open Market Operation	
Assets	Liabilities
Change in bond holdings: +\$1 million	Change in money stock: +\$1 million

QE = increase both asset and liabilities $\rightarrow i \downarrow$

QT = decrease - - - - -

4.2 Determining the Interest Rate: I (8 of 9)

- Suppose a bond such as a **Treasury bill**, or **T-bill**, promises to pay \$100 a year from now.
- If the price of the bond today is $\$P_B$, then the interest rate on the bond is:

$$i = \frac{\$100 - \$P_B}{\$P_B}$$

高利率會導致低的 price of bond

- *The higher the price of the bond, the lower the interest rate.*
- *The higher the interest rate, the lower the price today.*

4.2 Determining the Interest Rate: I (9 of 9)

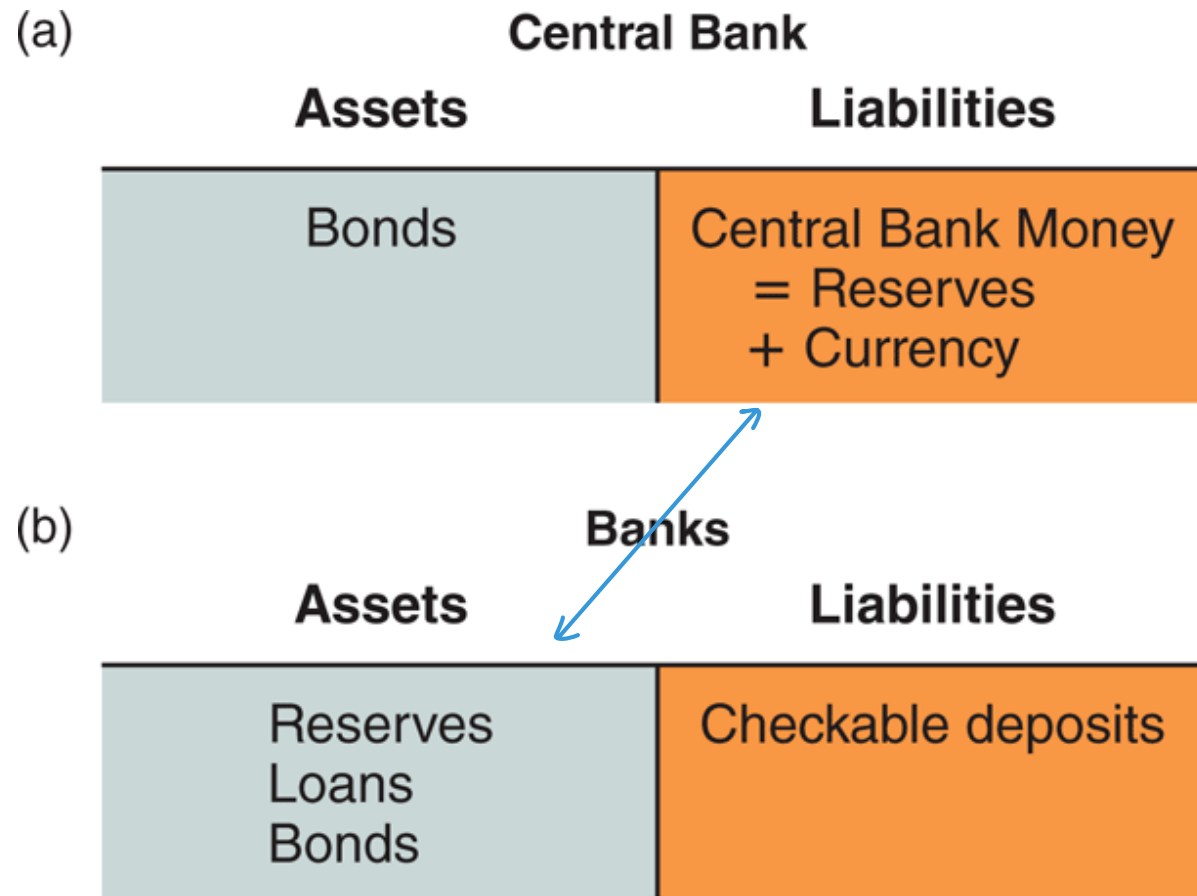
- Rather than the money supply, the central bank could have chosen the interest rate and then adjusted the money supply so as to achieve the interest rate it had chosen.
- Choosing the interest rate, instead of the money supply, is what modern central banks, including the Fed, typically do.

4.3 Determining the Interest Rate: II (1 of 6)

- **Financial intermediaries:** Institutions that receive funds from people and firms and use these funds to buy financial assets or to make loans to other people and firms.
- Banks are financial intermediaries that have money, in the form of *checkable deposits*, as their liabilities.
- Banks keep as **reserves** some of the funds they receive.
- The liabilities of the central bank are the money it has issued, called **central bank money**.

4.3 Determining the Interest Rate: II (2 of 6)

Figure 4.6 The Balance Sheet of Banks, and the Balance Sheet of the Central Bank Revisited



4.3 Determining the Interest Rate: II (3 of 6)

- Assume people hold **no currency** so the demand for money by people is the demand for checkable deposits:

$$M^d = \$Y L(i) \quad (4.3)$$

(—)

- The demand for reserves by banks depends on the amount of checkable deposits:

$$H^d = \theta M^d = \theta \$Y L(i) \quad (4.4)$$

- θ is the reserve ratio, and H^d is demand for high-power money or the monetary base.

$$H = \underset{\substack{|| \\ 0}}{C} + R$$

4.3 Determining the Interest Rate: II (4 of 6)

- Let H denote the supply of central bank money, then the equilibrium condition:

$$H = H^d \quad (4.5)$$

- Or using equation (4.4):

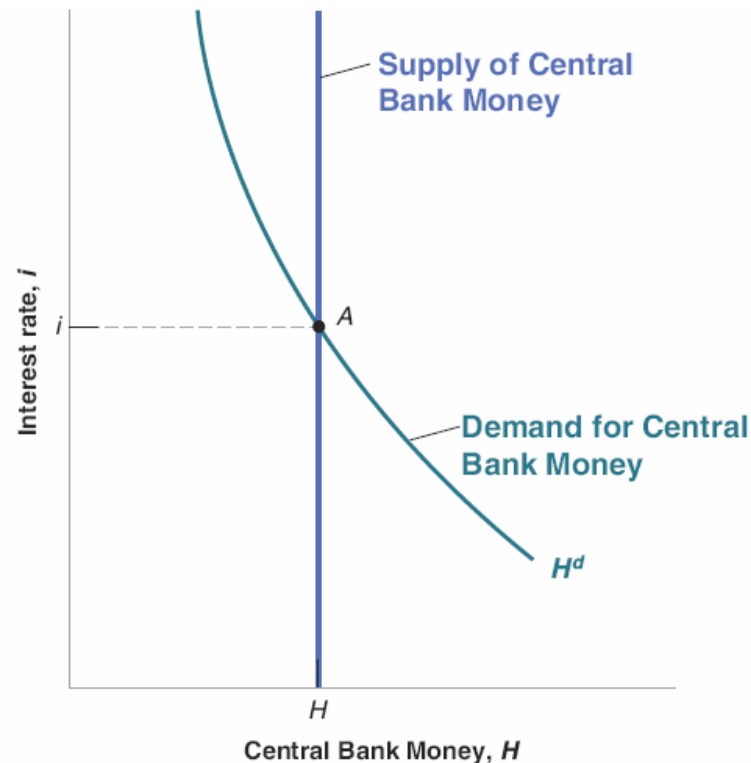
$$H = \theta \$Y L(i) \quad (4.6)$$

- An increase in H leads to a decrease in the interest rate, and a decrease in H leads to an increase in the interest rate.

4.3 Determining the Interest Rate: II (5 of 6)

Figure 4.7 Equilibrium in the Market for Central Bank Money and the Determination of the Interest Rate

The equilibrium interest rate is such that the supply of central bank money is equal to the demand for central bank money.



4.3 Determining the Interest Rate: II (6 of 6)

- The **federal funds market** is an actual market for bank reserves.
- The **federal funds rate** is the interest rate determined in the federal funds market.
- The federal funds rate is the main indicator of U.S. monetary policy because the Fed can choose the federal funds rate it wants by changing H .

$$H = \theta \$Y L(i)$$

FOCUS: Will Bitcoins Replace Dollars?

- Bitcoins are virtual assets that can be used for transactions.
- As of December 2018 the total value of bitcoin in circulation was \$67 billion.
- Bitcoins are not likely to replace dollars for 3 reasons:
 1. Most transactions are quoted in dollars, so price risk exists.
 2. Verifying bitcoin transactions takes too much electricity.
 3. Governments do not want bitcoin to be the accepted currency since they want to control monetary policy.

4.4 The Liquidity Trap (1 of 2)

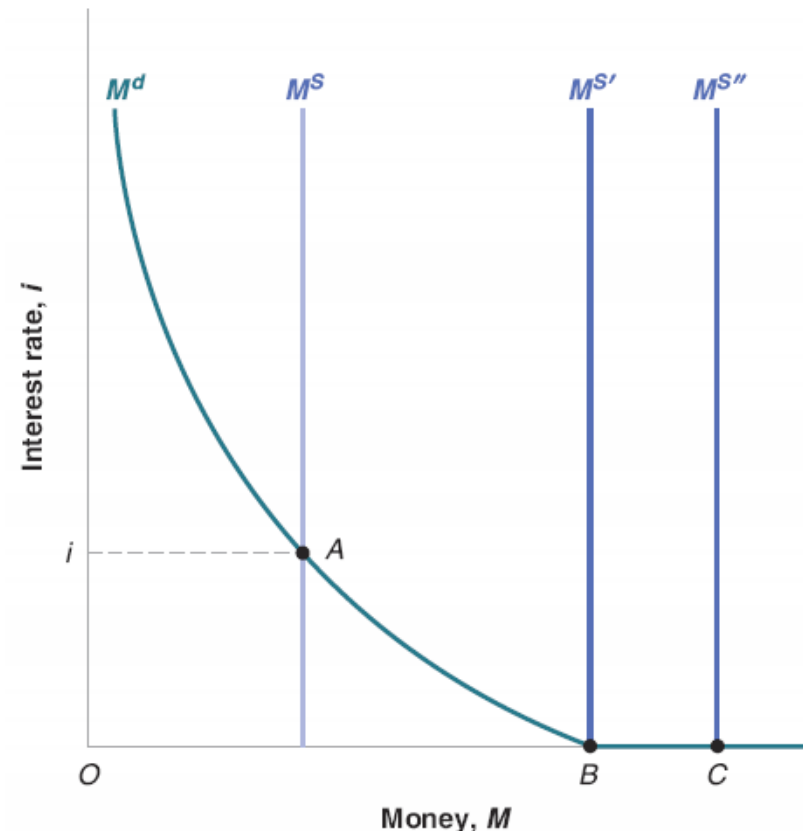
- **Zero lower bound:** The interest rate cannot go below zero.
- The economy is in a **liquidity trap** when the interest rate is down to zero, monetary policy cannot decrease it further.

4.4 The Liquidity Trap (2 of 2)

Figure 4.8 Money Demand, Money Supply, and the Liquidity Trap

When the interest rate is equal to zero, and once people have enough money for transaction purposes, they become indifferent between holding money and holding bonds. The demand for money becomes horizontal.

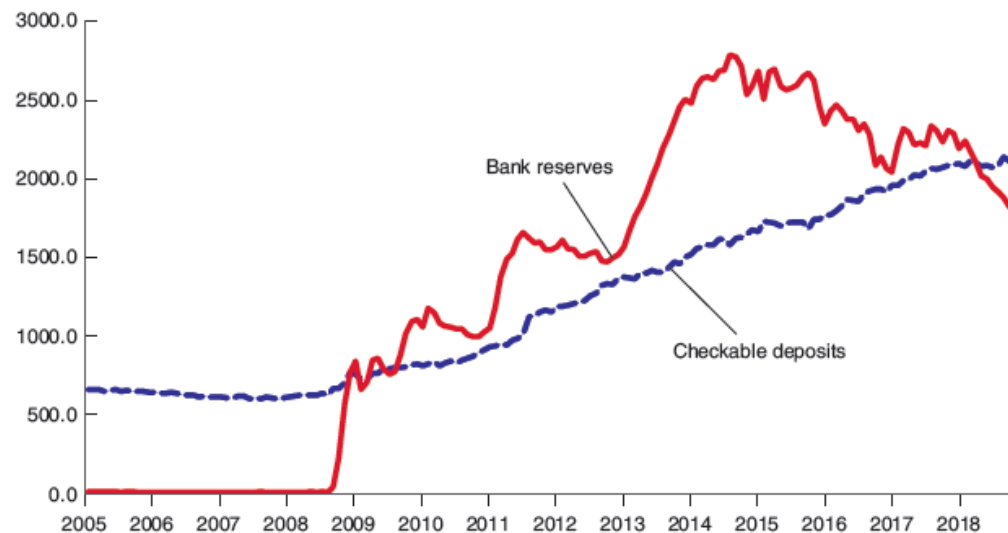
This implies that, when the interest rate is equal to zero, further increases in the money supply have no effect on the interest rate, which remains equal to zero.



FOCUS: The Liquidity Trap in Action

- The large increase in the supply of central bank money between 2008 and 2015 was absorbed by households and banks.

Figure 1 Checkable Deposits and Bank Reserves, 2005–2018 (billions).



Source: FRED: TCP, WRESBAL